

Estimating Nonlinear Insulin-Glucose Dynamics in Critically-ill Patients

Markus Mulvihill

KTH Royal Institute of Technology

September, 2026



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Thesis Presentation

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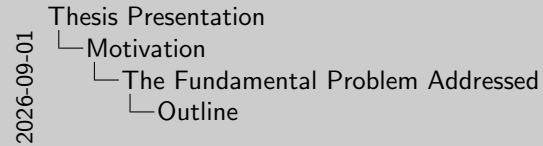
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1. Supervisor - Martin Lindström
2. Examiner - Ragnar Thobaben
3. Opponent - Divyayan Day

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 - The Fundamental Problem Addressed
 - Background
 - Previous Work
- 2 Results and Contributions
 - Main Results
 - Key Ideas Behind Implementation
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Fundamentals of Glucoregulation

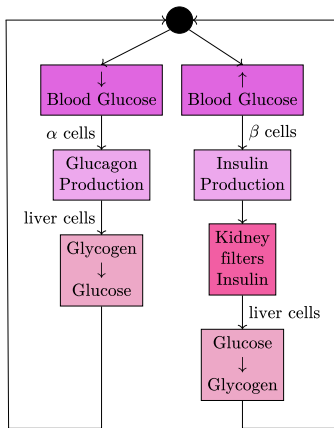


Figure: Flowchart of glucoregulation in healthy individuals.

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Fundamentals of Glucoregulation



Figure: Flowchart of glucoregulation in healthy individuals.

1. The thesis revolves around the functionality of glucoregulation.
2. A high level diagram of the human insulin-glucose system can be described as the image.

Insulin Resistance in ICU Patients

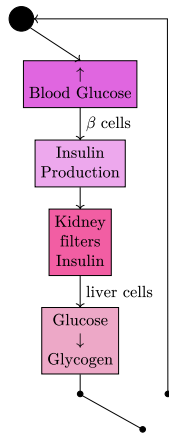


Figure: Consequence of insulin resistance in ICU patients.

- Critically ill patients' bodies do not respond normally to insulin.
- Stress induced from critical illness has been observed with an increase of resistance to insulin.
- Impaired glycogen production in the liver creates an excess of blood glucose (hyperglycaemia).

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Insulin Resistance in ICU Patients

Insulin Resistance in ICU Patients

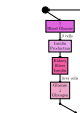


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- Critically ill patients' bodies do not respond normally to insulin.
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Mention that the break in the chain is just for visual purposes.

Intensive Insulin Therapy

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-Intensive Insulin Therapy

- **What:** Utilizes aggressive insulin infusions and feeding inputs to tightly control blood glucose.
- **Goal:** Prevents hyperglycaemia in patients to reduce **mortalities** and longer visitation stays.
- **How:** Traditionally has relied on ad-hoc sliding protocols.

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Overview of Ad-hoc Sliding Protocols

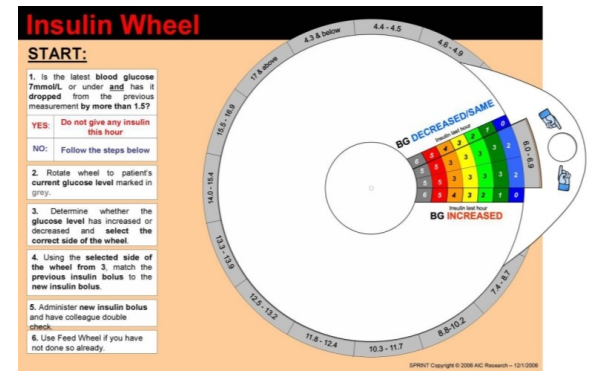


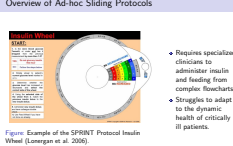
Figure: Example of the SPRINT Protocol Insulin Wheel (Loneragan et al. 2006).

- Requires specialized clinicians to administer insulin and feeding from complex flowcharts.
- Struggles to adapt to the dynamic health of critically ill patients.

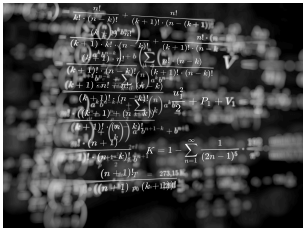
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- └ Overview of Ad-hoc Sliding Protocols



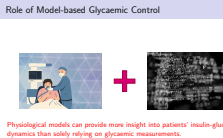
Role of Model-based Glycaemic Control



Physiological models can provide more insight into patients' insulin-glucose dynamics than solely relying on glycaemic measurements.

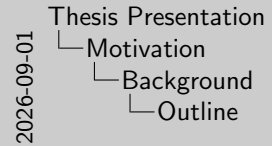
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Insulin infusions combined with better understanding of dynamics can provide more insight to the clinicians

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Intensive Control Insulin-Nutrition-Glucose (ICING) model

Lin et al. has developed a clinically-validated nonlinear insulin-glucose model with the following main components

Dynamic	Notation	Value
Blood Glucose	$\dot{G}$	mmol/L
Interstitial Plasma	$\dot{Q}$	mU/L
Plasma Insulin	$\dot{i}$	mU/L
Glucose in the Stomach	$\dot{P}_1$	mmol
Glucose in the Gut	$\dot{P}_2$	mmol
Endogenous Insulin Production	$u_{en}(t)$	mU/min
Insulin Sensitivity	$S_I(t)$	-
Input	Notation	Value
Exogenous Insulin	$u_{ex}(t)$	mU/min
Entenal Food	$D(t)$	mmol/min
Parenteral Glucose	$P_N(t)$	mmol/min

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Computational Approximation of the ICING model

- The ICING model equations are defined in continuous-time
- Discretization of the model is required for the presented estimation methods

For an example dynamic F

$$\begin{aligned}t_k &= t_{k-1} + \Delta, \\ \dot{F} &\approx \frac{F(t_k) - F(t_k - \Delta)}{\Delta}, \\ F_k &\approx F_{k-1} + \Delta \dot{F}.\end{aligned}$$

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1. According to Euler's method
2. Discretization error is bounded by Δ

The discrete-time dynamic ICING model can be represented in the stochastic state-space.

State-space Representation

$$x_k \sim p(x_k \mid x_{k-1}, u_{k-1})$$
$$y_k \sim p(y_k \mid x_k)$$

Parameter Description

System Variable	Notation
Dynamics	x_k
Inputs	u_k
Measurements	y_k

There are various methods of optimal filtering, and each has their own assumptions and limitations

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 - Bayesian Filtering

Bayesian Filtering

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There are various methods of optimal filtering, and each has their own assumptions and limitations

1. The measurement often depends on the hospital, but will be discussed more later
2. Just the EKF and the PF will be in the topic of this thesis

Extended Kalman Filter

Approximates **nonlinear** and **Gaussian** dynamic and measurement models as a linear transformation.

1 Prediction step

$$m_k^- = f(m_{k-1}, u_{k-1}),$$

$$P_k^- = F_x(m_{k-1})P_{k-1}F_x^T(m_{k-1}) + F_q(m_{k-1})Q_{k-1}F_q^T(m_{k-1}).$$

2 Update step

$$v_k = y_k - h(m_k^-),$$

$$S_k = H_x(m_k^-)P_k^-H_x^T(m_k^-) + H_r(m_k^-)R_kH_r^T(m_k^-),$$

$$K_k = P_k^-H_k^T(m_k^-)S_k^{-1},$$

$$m_k = m_k^- + K_k v_k,$$

$$P_k = P_k^- - K_k S_k K_k^T.$$

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Bootstrap Particle Filter

Formulates the posterior of the dynamic function as a Monte Carlo approximation.

- 1 Sample N particles from the dynamic function

$$x_k^{(i)} \sim p(x_k | x_{k-1}^{(i)}, u_{k-1}) \quad i = 1, \dots, N.$$

- 2 Update weights from the measurement

$$w_k^{*(i)} \propto p(y_k | x_k^{(i)}) \quad i = 1, \dots, N,$$

$$w_k^{(i)} = \frac{w_k^{*(i)}}{\sum_{j=1}^N w_k^{*(j)}} \quad i = 1, \dots, N.$$

- 3 Systematically resample particles from a distribution proportional to the weights

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Formulation of Synthetic Patients

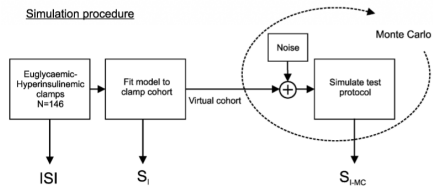


Figure: An example of a simulation procedure (Lotz et al. 2008).

- Synthetic patient trajectories are utilized to model clinical insulin-glucose data.
- Such data are often limited to perturbations in previously recorded clinical data.
- Sampling from a distribution of patient insulin-glucose states would allow for a greater number of diverse synthetic trajectories.

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Estimation of Patient Insulin-Glucose Dynamics

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Motivation

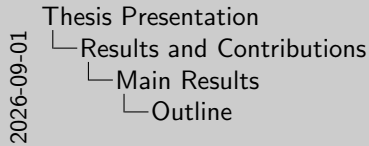
Previous Work

Estimation of Patient Insulin-Glucose Dynamics

- Previous studies have employed Bayesian filtering methods:
 - Extended and unscented Kalman filters
 - Particle filters
- Practical implementation is often insufficiently addressed:
 - Constant nutrition and insulin inputs
 - Known initial insulin-glucose dynamics
- Estimator configurations and performance are not systematically compared.

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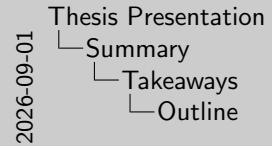
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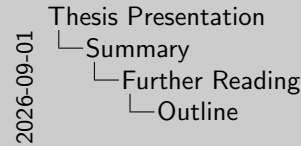
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References

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2. Lotz, Thomas F. et al. (Mar. 2008). "Monte Carlo analysis of a new model-based method for insulin sensitivity testing". In. DOI: 10.1016/j.cmpb.2007.03.007.

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